Adil Anwar Sadath

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.41
National Higher Secondary School, Kolathur	Class XII (Kerala HSE)	2022	99.17%
Delhi International School, Valanchery	Class X (CBSE)	2020	93.40%

PROJECTS

- **Automation of CAD and CFD to generate datasets for a prototype-specific ML model :**

Engineered an automated *SolidWorks-Ansys* Fluent workflow to generate aerodynamic dataset for a hypersonic glide vehicle and trained a CNN model to predict lift and drag directly, reducing the need for computationally intensive CFD simulations. *(DRDO Project, Prof. L. D. Chandrala)*

- **Calibration of Fracture loci for heat-treated steels and training an ML model to predict fracture loci:**

Conducted experimental testing and *ABAQUS* simulations on various in-house heat-treated steel specimens to obtain fracture-point parameters, and trained an ML model, significantly reducing the computational cost and engineering complexity associated with fracture models.

Used the ML model inference to quantify the influence of strain rate, heat treatment, and chemical composition on the fracture behavior of steel. *(Prof. S.N. Khaderi)*

- **Design and Fabrication of a 360 V Battery Pack for a Race Car:**

Designed the structural casing and mounting, thermal management, and HV electrical integration for a 360 V battery accumulator, and conducted DFMEA for design risk assessment, complying with Formula Student Germany *(FSG)* safety constraints. *(Formula Student Electric)*

- **Coin-Operated Medicine Dispenser with Variable Dosage Mechanism**

Designed and fabricated a fully mechanical coin-actuated gear-shifting mechanism to vary the transmission ratio and control the dosage delivered by a peristaltic pump based on the coin size. *(Departmental Project)*

- **Design and Fabrication of a *GFRP* Air-Intake Manifold:**

Performed CFD and composite structural analysis, and fabricated a hand-laid glass fiber intake manifold for a high-performance FSAE race car engine. *(Formula Student Combustion)*

- **Research & Development of a Scored Rupture Disc for a Single-Stage Light Gas Gun (SSLGG):**

Applied calibrated fracture parameters from my fracture studies to the design and analysis of a scored rupture disc, for the manufacturing of an SSLGG for impact dynamics research. *(Prof. S.N. Khaderi)*

- **CAD Design and Assembly of a Coupler Torque-testing Rig:**

Designed a hydraulic-actuated torque-testing rig for mechanical couplers, replacing a high-torque motor with a hydraulic actuator to reduce cost while maintaining the required testing torque. *(Machine Design Course Project)*

- **Design for Manufacturing & Assembly Optimization -- Formula Student**

Provided DFM/DFA guidance across the vehicle design team, optimizing uprights, wheel hubs, chassis, mounts, and other components and reduced manufacturing cost and material waste while adapting designs to available machining processes, contributing to design reviews and manufacturing decisions across multiple subsystems.

- **Noise Control of a Portable Blender**

Implemented noise control strategies, achieving a 7 dB reduction in overall sound level using vibration-damping sheets and acoustic foam. *(Noise Control Course Project)*

SKILLS

- **FEA and CFD**: Ansys Fluent, Ansys Mechanical, ABAQUS
- **CAD and Drafting**: SolidWorks, Fusion 360, Solid Edge, Ansys SpaceClaim
- **Machine Learning | Python | MATLAB | Simulink**
- **Design and Manufacturing**: GD&T, DFM/DFA, DFMEA

POSITIONS OF RESPONSIBILITY

- **Team Captain**, Pravah Racing 2025-26
- **Vice Captain,** Aeolus Racing 2024-25
- **Teaching Assistant,** Introduction to Programming course 2024
- **Head of Multimedia,** Finance and Consulting Club 2025-26
- **Coordinator**, International Relations Cell 2024-25
- **Hospitality and Logistics coordinator,** Department Day 2025

RELEVANT COURSES

- Vehicle Dynamics | Kinematics and Dynamics | Machine Design | Solids and Fluids
- Machine learning | Control systems | Marine Robotics
- Spoken Japanese Basics

EXTRACURRICULAR

- Participant in INTER-IIT CULTURALS for **Filmmaking**
- National Sports Organization - **Football Player**